LeSuccess Academy

Learn . Educate . Succeed | Official Course Syllabus

Data Science

Duration: 3-6 Months | Mode: HYBRID | 100% Placement Support

Detailed Curriculum & Module Outline:

Module 1: Python for Data Science

Python fundamentals aimed squarely at data work - collections, functions, files and Jupyter

Module 2: NumPy and Pandas

Arrays, dataframes, joins, group-bys and reshaping, on datasets too large to eyeball

Module 3: SQL for Data Science

Extracting and aggregating from relational databases, including window functions and query perf

Module 4: Statistics and Probability

Distributions, sampling, confidence intervals and hypothesis testing, applied to real questions

Module 5: Exploratory Data Analysis

Matplotlib and Seaborn, outliers, correlations and the plots that change what you build next

Module 6: Data Cleaning and Feature Engineering

Missing values, encoding, scaling and features that add signal rather than noise

Module 7: Supervised Learning: Regression

Linear and regularised regression, residuals, and reading what a model is telling you

Module 8: Supervised Learning: Classification

Logistic regression, decision trees, random forests and gradient boosting on imbalanced data

Module 9: Unsupervised Learning

K-means, hierarchical clustering and dimensionality reduction, with the hard problem of validat

Module 10: Model Evaluation and Tuning

Train/test splits, cross-validation, precision and recall, overfitting and hyperparameter searc

Module 11: Model Deployment Basics

Wrapping a model in an API, versioning it, and what breaks once it meets live data

Module 12: Capstone Data Science Project

An end-to-end project from question to deployed model, reviewed and presented